Reader: We have already noted one of such theorems: the theorem on the uniqueness of the limit of function at a point.

Author: This theorem is analogous to that about the uniqueness of the limit of numerical sequence.
I shall also give (without proof) the theorems on the limit of the sum, the product, and the ratio of functions.

Theorems: If functions $f(x)$ and $g(x)$ have limits at a point a, then functions $[f(x)+g(x)]$, $[f(x)g(x)]$, $\left(\frac{f(x)}{g(x)}\right)$ also have limits at this point. These limits equal the sum, product, and ratio, respectively, of the limits of the constituent functions (in the last case it is necessary that the limit of the function $g(x)$ at a be different from zero).

Thus $\boxed{\lim_{x\to a}[f(x)+g(x)] = \lim_{x\to a} f(x) + \lim_{x\to a} g(x)}$

$$\boxed{\lim_{x\to a}[f(x)\,g(x)] = \lim_{x\to a} f(x)\,\lim_{x\to a} g(x)}$$

$$\lim_{x\to a}\left(\frac{f(x)}{g(x)}\right) = \frac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)}$$

under an additional condition: $\boxed{\lim_{x\to a} g(x) \neq 0}$

Reader: We have already discussed the similar theorems for numerical sequences.

Author: Next I wish to make two remarks, using for the purpose specially selected examples.

Note 1: It is illustrated by the following example. Obviously $\lim_{x\to 1}\sqrt{1-x^2}=0$ and $\lim_{x\to 1}\sqrt{x-1}=0$. Does it mean that

$$\boxed{\lim_{x\to 1}\left(\sqrt{1-x^2}+\sqrt{x-1}\right)=0\,?}$$